

US Healthcare Current State – Evidence Gap (AI in Healthcare):

While healthcare data is now abundant, it remains siloed and difficult to transform into actionable evidence. Today, only 14% of daily medical decisions are based on high-quality evidence—a disconnect known as “The Evidence Gap.” Traditional clinical trials are limited and exclude nearly 70% of real-world patients, including those with comorbidities, as well as underrepresented populations such as minorities, children, and seniors, among others.

Generating high-quality insights typically takes months and significant resources, confining the process to expensive, academic research settings. The current system places significant strain on U.S. healthcare: many patients do not receive evidence-based care, leading to avoidable adverse outcomes, under or delayed treatments, and increased costs.

About Atropos Health:

Atropos Health is on a mission to make personalized evidence available to every patient. We achieve this mission starting with our GENEVA OS™ platform that automates the generation of publication-grade, observational research studies for physicians, researchers, and patients, closing the evidence gap from bench to bedside. We partner with leading health systems, life sciences organizations, pharmaceutical companies, and insurers to improve healthcare across 40+ specialties in the USA.

The Atropos Evidence™ Network, is the largest federated data network in healthcare with over 300 million patient records spanning electronic medical record, claims, lab, and specialty data. As Health Systems continue to drive policy and standards to enhance data infrastructure and interoperability, Atropos Health's capabilities are a prime example of how data can be used to create actionable insights and improve decision making at both the point of care and lead to life saving drug discoveries with speed and accuracy.

Specific Questions

In addition to the general requests for information above regarding AI regulation, reimbursement, and research & development, HHS seeks input on the following specific questions:

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

The biggest barriers to private sector AI innovation and adoption in clinical care are regulatory and liability uncertainty, slow and fragmented health system governance, and lack of standardized evaluation and procurement pathways. These are compounded by difficult EHR integration, data access and interoperability constraints.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

For non-medical device AI used in clinical care, novel legal and implementation challenges include unclear liability and indemnification when AI-informed decisions contribute to harm, ambiguity around accountability between clinicians, health systems, and vendors, and evolving privacy and security risks associated with large-scale data use and model updating. These challenges strain existing governance structures and contribute to conservative adoption. HHS can play a critical role by clarifying liability and accountability expectations, issuing guidance on privacy, security, and responsible data use for clinical AI, and promoting standardized governance and contracting frameworks that reduce risk and administrative friction while protecting patients.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

For non-medical device AI in clinical care, the most effective evaluation combines strong pre-deployment validation with continuous post-deployment monitoring and human-centered workflow assessment. Before go-live, models should be tested across sites and subpopulations, evaluated in silent trials, assessed for usability and workflow fit, and stress-tested for robustness to data shifts. After deployment, ongoing monitoring for performance drift, safety, and real-world clinical and operational impact is essential. Evaluation should focus not only on accuracy, but also calibration, clinical utility, equity, and clinician interaction. HHS support would meaningfully accelerate safe and scalable adoption by establishing shared evaluation infrastructure and standards, with cooperative agreements, targeted grants, and reference implementations as the most impactful mechanisms.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

The greatest future impact will come from actionable, transparent decision support grounded in real-world evidence at the point of care; multimodal tools that improve high-stakes medication decisions and prevent adverse events; and AI that surfaces care gaps and quality and equity insights in real time. These approaches offer the strongest path to improving outcomes and quality while reducing avoidable utilization and clinician burden. Atropos has proven this at scale working with leading health system organizations across the US.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

The adoption of AI for clinical care is most strongly influenced by clinical leadership (CMO, CMIO, CNIO, department chairs), who determine clinical relevance and frontline trust; executive leadership (CEO, COO, CFO), who set strategic priorities, risk tolerance, and funding; IT and AI governance bodies (CIO, digital health and AI committees), who control integration, security, and scalability; and compliance, legal, and quality leadership, who shape regulatory interpretation, liability posture, and patient safety standards. The primary administrative hurdles include fragmented governance and slow, multi-layered approval processes; regulatory and liability uncertainty around clinical AI; integration challenges with EHRs and clinical workflows; lack of standardized evidence and evaluation frameworks to support purchasing decisions; data governance and interoperability constraints; and procurement and contracting processes that are misaligned with the pace of AI innovation.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Enhanced interoperability would materially expand market opportunities, accelerate innovation, and improve the real-world impact of AI in clinical care by enabling timely, standardized access to high-quality longitudinal data across settings and populations. The greatest impact would come from broader adoption of common data standards for structured EHR data to support generalizable real-world evidence; improved linkage of clinical, utilization, pharmacy, claims, and patient-reported outcomes data across care settings to assess downstream outcomes, safety, and cost; and the development of shared benchmarking and evaluation frameworks so AI tools can be consistently validated and trusted across institutions and populations..

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Patients and caregivers want AI in clinical care to address persistent challenges such as variation in care, delays in diagnosis and treatment, lack of personalization, and limited

transparency in clinical decision-making. AI tools that leverage real-world clinical data can help clinicians make more consistent, timely, and patient-specific decisions by showing how similar patients have been treated and what outcomes were achieved.

At the same time, patients and caregivers express concerns about trust, bias, and the appropriate role of AI in care. They are wary of AI systems that appear to replace clinician judgment, lack transparency, or are trained on non-representative data. Patients also emphasize the importance of data privacy, equity, and clear accountability for clinical decisions.

Overall, patients and caregivers are most supportive of AI that augments clinician decision-making with interpretable, real-world evidence, improves equity and consistency of care, and operates within strong governance and privacy safeguards.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

- a. Are there published findings about the impact of adopted AI tools and their use clinical care?
- b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?